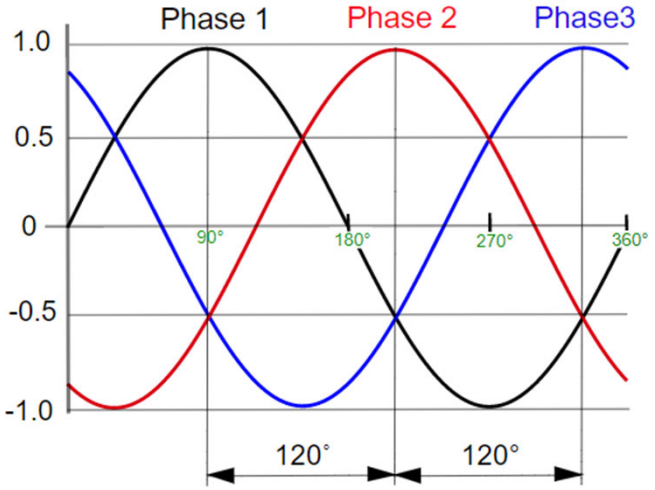
ECE 330

Chapter 2

Phasors and three-phase circuits



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University of Illinois at Urbana-Champaign
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Learning Objectives: you will be able to

- Convert time-domain sinusoidal sources to phasor form using **RMS values**
- Use **RMS phasors** to analyze steady-state AC circuits
- Define and calculate **real (P), reactive (Q)**, **apparent (S)** and **complex power ($\bar{S}$)**
- Define **power factor (PF),** and classify loads as having **leading, lagging, or unity** power factor.
- Use the **power triangle** to compute power and power factor.
- Explain the motivation for **power factor correction** in AC power systems, and calculate the required reactive compensation to achieve a desired PF.
- Explain the motivation for using three-phase power systems.
- Identify **Wye (Y)** and **Delta (Δ)** three-phase connections and analyze balanced three-phase loads connected in Y or Δ configurations.
- Develop **per-phase equivalent circuits** for balanced Y- and Δ-connected loads and use single-phase circuit analysis techniques to solve three-phase power problems

Review of Phasors

- ECE 210

$$A(t) = A_m \cos(\omega t + \theta)$$

- ECE 330

$$A(t) = A_m \cos(\omega t + \theta)$$

Converting from sine to cosine

$$v(t) = \sqrt{2}(10)\sin\left(wt + 30^\circ\right)$$

$$i(t) = -\sqrt{2}(5)\sin\left(wt - 20^\circ\right)$$

Example 1: write as phasors

$$i(t) = \sqrt{2}(5)\cos\left(wt - 20^\circ\right)$$

A) $\bar{I} = \sqrt{2}(5)\angle -20^\circ$

B) $\bar{I} = (5)\angle -110^\circ$

C) $\bar{I} = \sqrt{2}(5)\angle -110^\circ$

D) $\bar{I} = (5)\angle -20^\circ$

Example 2: write as phasors

$$i(t) = -\sqrt{2}(5)\sin\left(wt + 30^\circ\right)$$

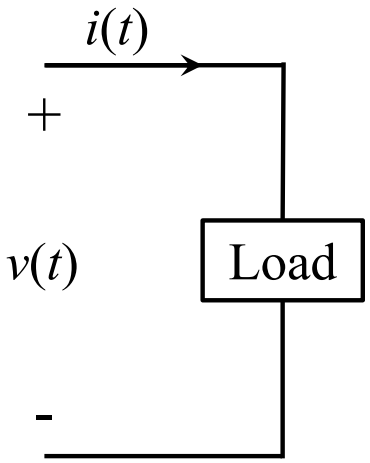
A) $\bar{I} = (5)\angle 120^\circ$

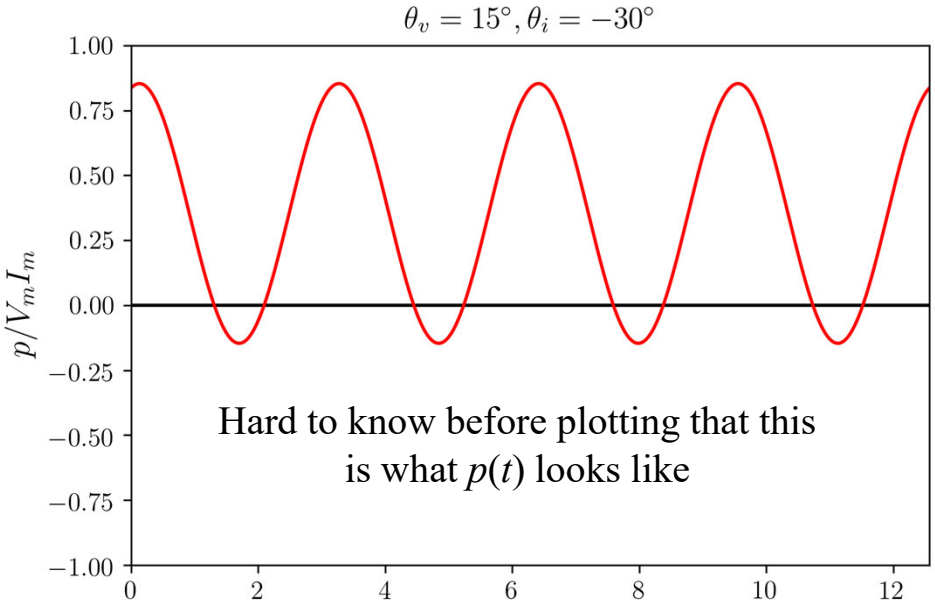
B) $\bar{I} = \sqrt{2}(5)\angle 30^\circ$

C) $\bar{I} = \sqrt{2}(5)\angle 120^\circ$

D) $\bar{I} = (5)\angle 30^\circ$

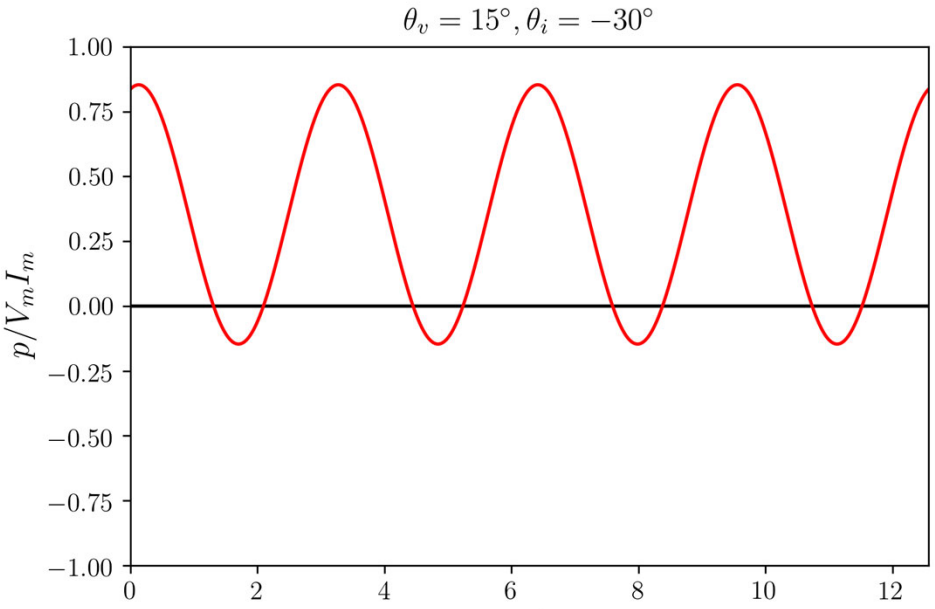
Single Phase: 1 voltage and 1 current





Simplifying p(t)

$$\cos(A)\cos(B) = \frac{1}{2}\left[\cos(A+B) + \cos(A-B)\right]$$



Average Power: *P*

Reactive (Q) and Complex ($\bar{S}$) power

Example 3:

$\bar{V} = 10\angle 15^\circ$ V, $\bar{I} = 5\angle -30^\circ$ A

Find P :

Example 4:

$\bar{V} = 10\angle 15^\circ$ V, $\bar{I} = 5\angle -30^\circ$ A

Find Q:

A) 50 W

B) 35.355 W

C) -12.94 VAR

D) 35.355 VAR

Example 5:

$$v(t) = \sqrt{2}(10)\cos\left(wt + 15^\circ\right)$$

$$i(t) = \sqrt{2}(7)\sin\left(\omega t + 75^\circ\right)$$

Find $\overline{S}$:

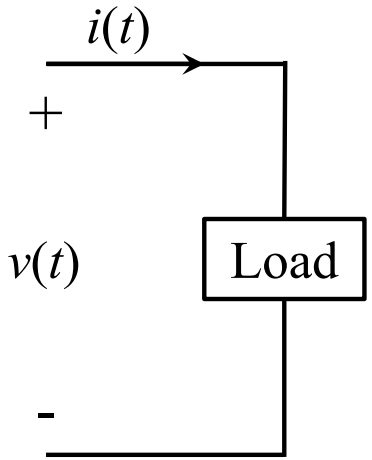
A) $70+j0$ VA

B) $35-j60.62$ VA

C) $60.62+j35$ VA

D) $121.24+j70$ VA

Another way to view complex power

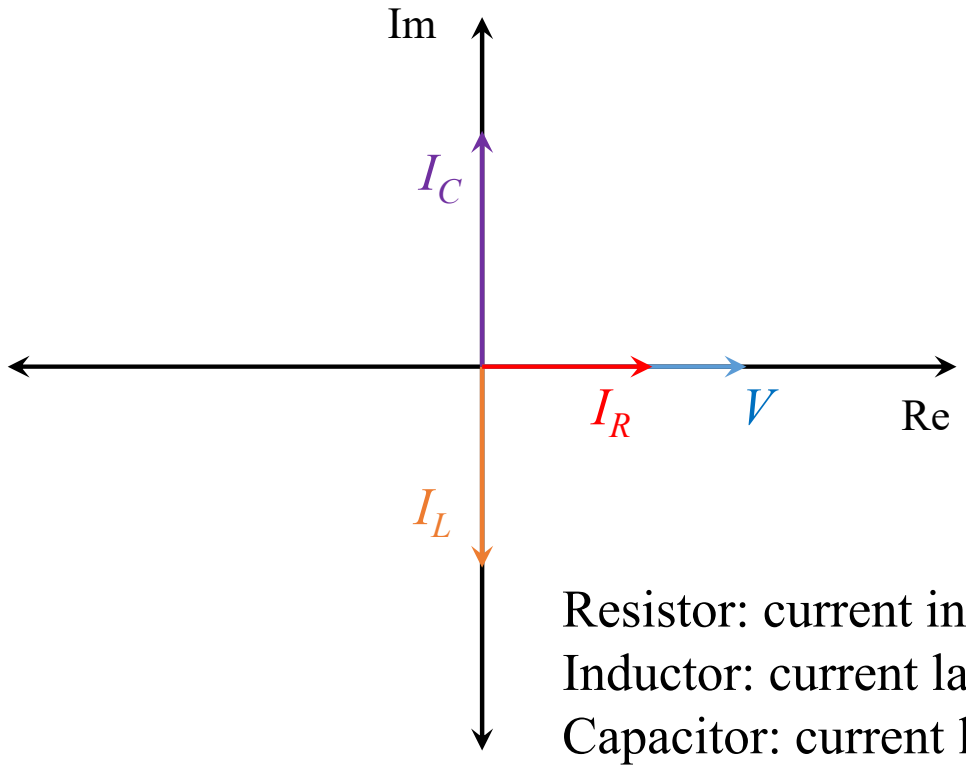


Recall: cosine is an even function

$$\cos(x)=\cos(-x)$$

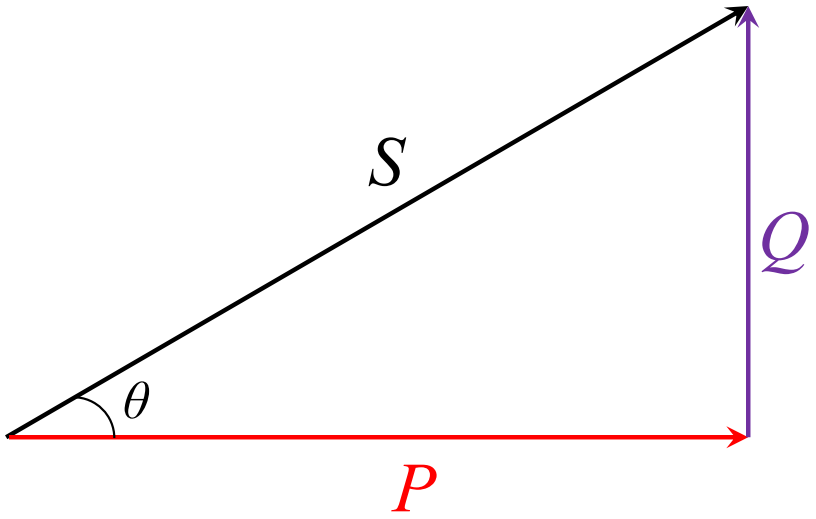
How do leading and lagging PF correspond to circuit?

Plotting in Complex Plane

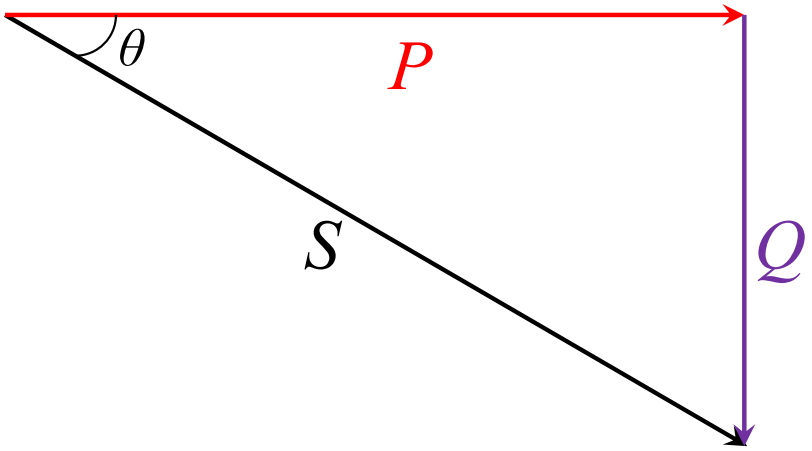


Resistor: current in phase with voltage
Inductor: current lags voltage by 90°
Capacitor: current leads voltage by 90°

Power Triangle: Lagging PF



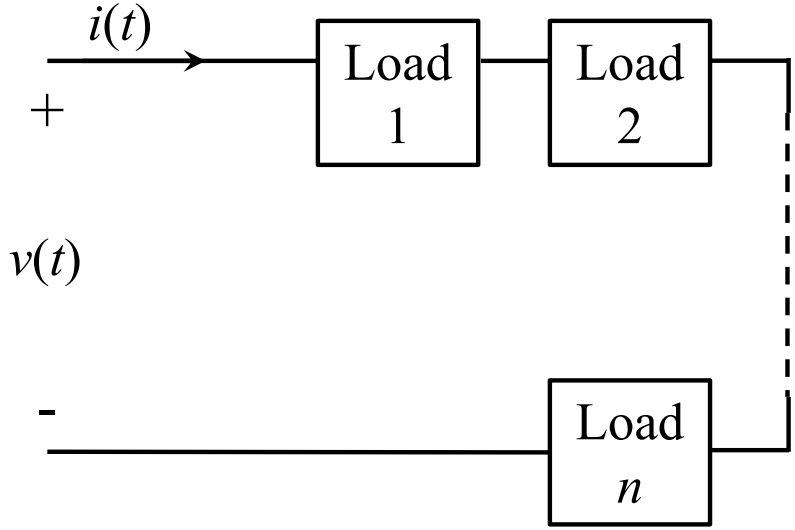
Power Triangle: Leading PF

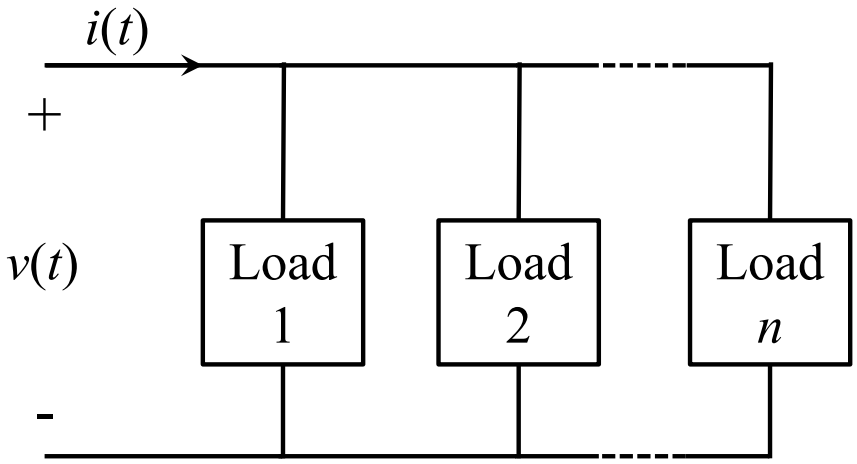


Specifying Power

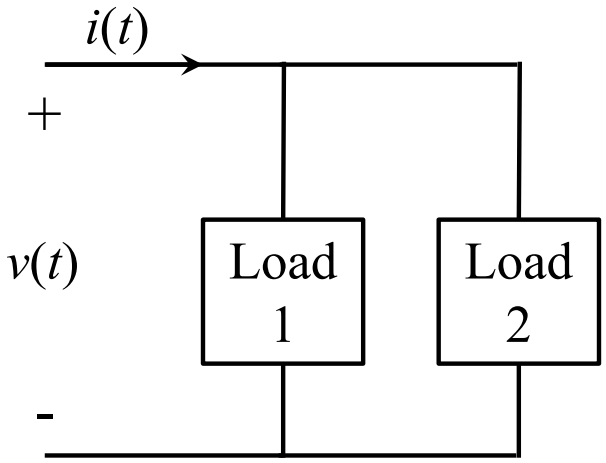
- 6 total quantities:
- Need a reference: assume
- Use whatever is given to calculate unknown quantities

What if there are multiple loads?





Example 6:



$$\bar{V} = 100\angle 10^\circ \text{ V}$$

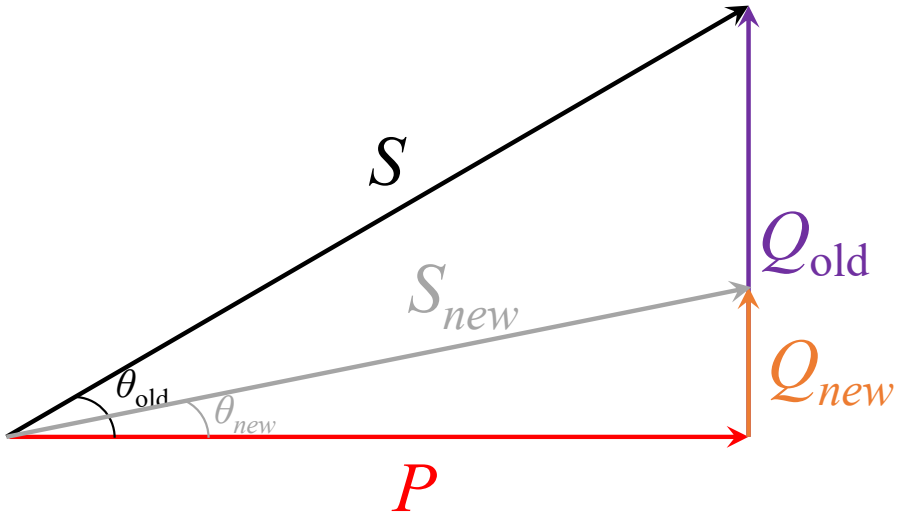
$$\bar{I}_1 = 10\angle -26.8^\circ \text{ A}$$

$$\bar{I}_2 = 5\angle 40^\circ \text{ A}$$

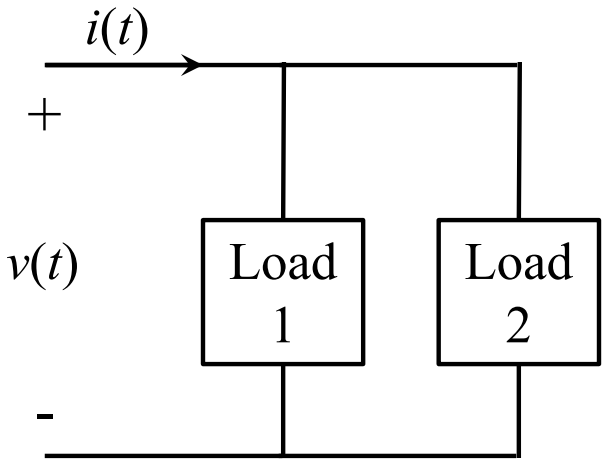
Purely reactive loads:

Why PF correction?

Power Triangle: Power Factor Correction



Example 7: PF correction



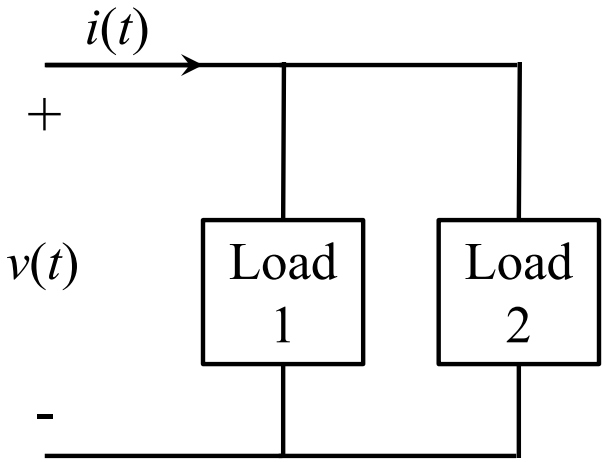
What should capacitor in parallel add to get PF = 0.98 lagging?

$$\bar{V} = 100\angle 10^\circ \text{ V}$$

$$\bar{I}_1 = 10\angle -26.8^\circ \text{ A}$$

$$\bar{I}_2 = 5\angle 40^\circ \text{ A}$$

Example 8: PF correction



$\bar{V} = 100\angle 10^\circ$ V

$\bar{I}_1 = 10\angle -26.8^\circ$ A

$\bar{I}_2 = 5\angle 40^\circ$ A

What should capacitor in parallel add to get PF=1?

A) 0 VAR of capacitance

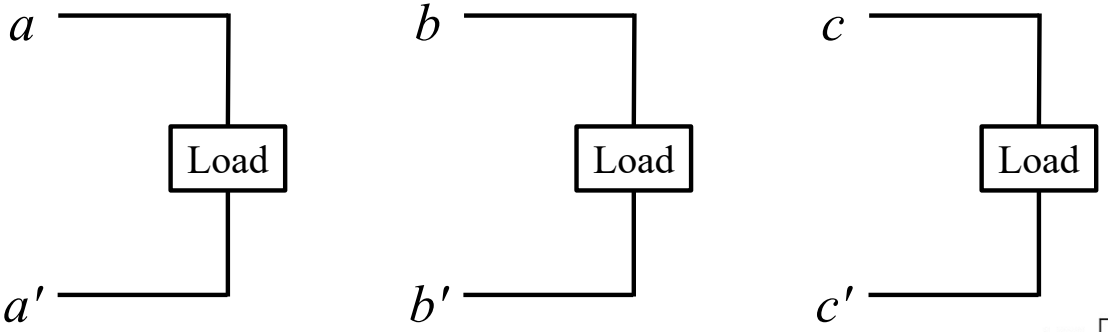
B) 600 VAR of capacitance

C) 100 VAR of capacitance

D) 350 VAR of capacitance

2 Phase system

3 phase power



Positive Sequence:

a - b - c

$$V_{aa'} = V_m \cos(\omega t)$$

$$V_{bb'} = V_m \cos(\omega t - 120^\circ)$$

$$V_{cc'} = V_m \cos(\omega t + 120^\circ)$$

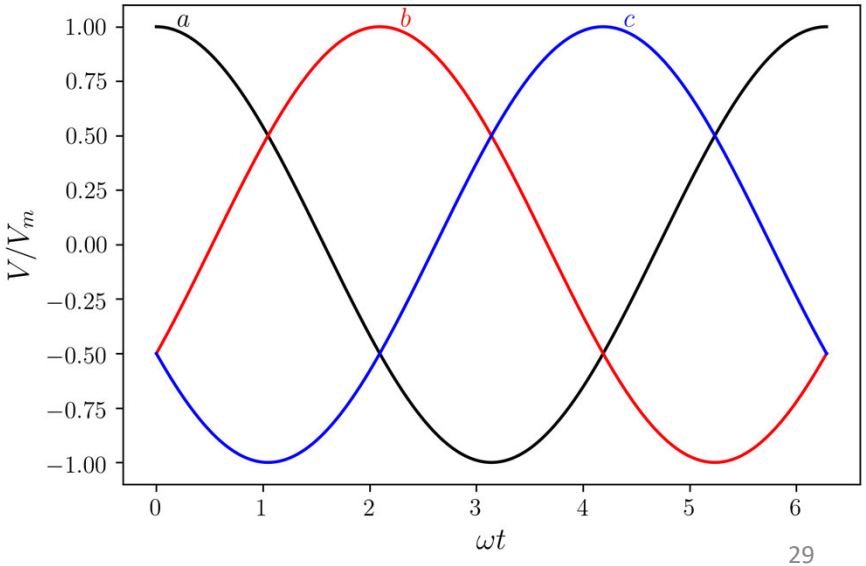
Negative Sequence:

a - c - b

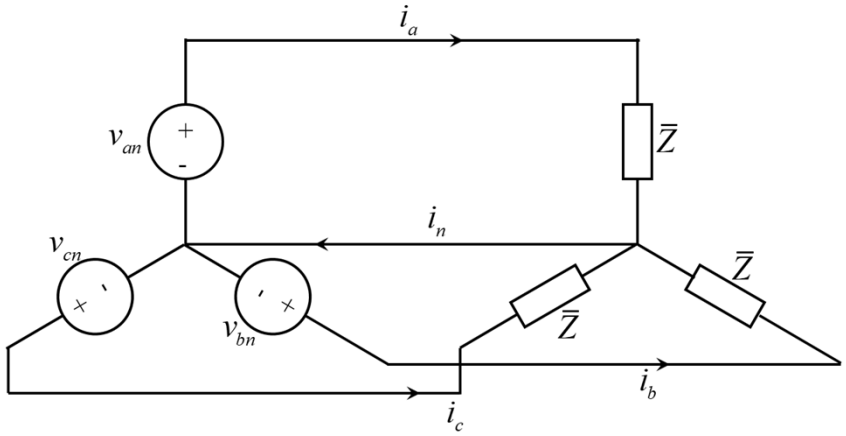
$$V_{aa'} = V_m \cos(\omega t)$$

$$V_{bb'} = V_m \cos(\omega t + 120^\circ)$$

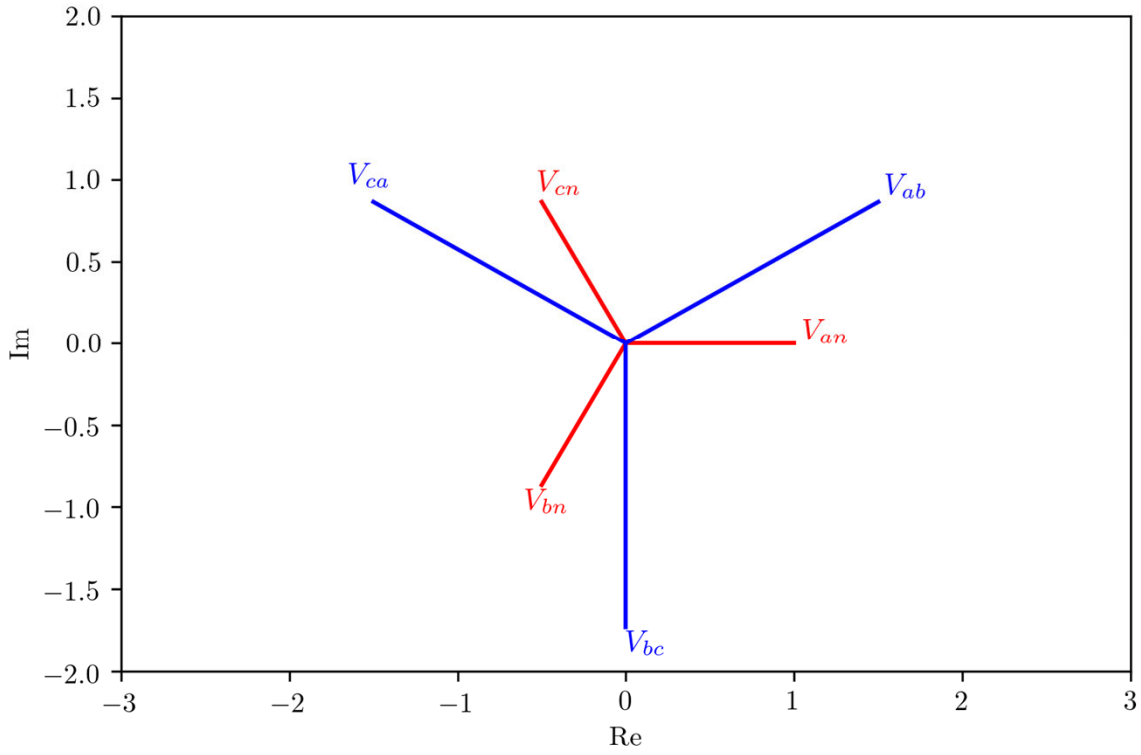
$$V_{cc'} = V_m \cos(\omega t - 120^\circ)$$



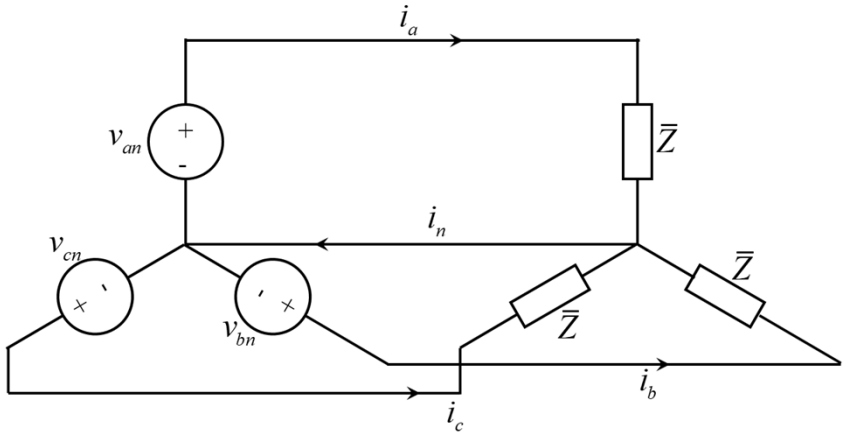
Wye-Wye Connection



Phasor Diagram



Currents in each phase/line



Complex Power

Example 9:

Wye-connected load, $V_L = 480\ V, P = 24\ kW, PF = 0.8\ lag.$
Find I_L

Example 9-cont:

Wye-connected load, $V_L = 480\ V, P = 24\ kW, PF = 0.8\ lag.$
Find I_L

Wye-connected load, $V_L = 480\ V, P = 24\ kW, PF = 0.8\ lag.$
Find $\bar{S}$

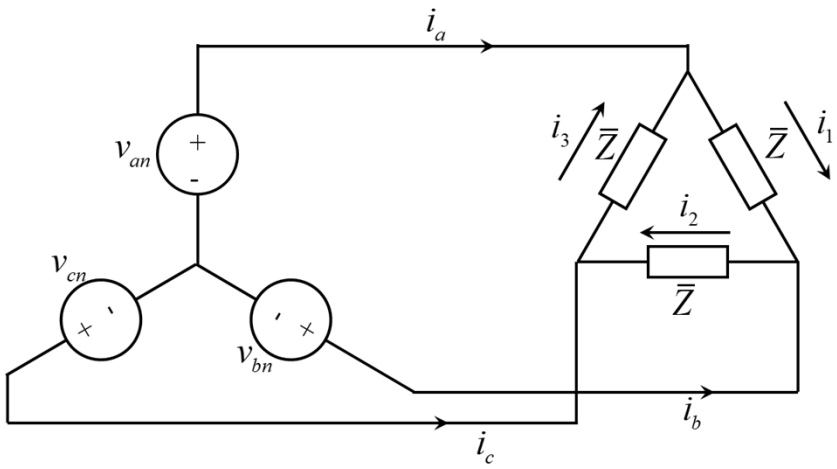
A) $17318.4\angle 36.87^\circ$ VA

B) $30000\angle 36.87^\circ$ VA

C) $30000\angle -36.87^\circ$ VA

D) $17318.4\angle -36.87^\circ$ VA

Wye-Delta Connection



Complex Power

Example 10:

Delta-connected load, $V_L = 480\ V, P = 24\ kW, PF = 0.8\ lag.$
Find I_L

A) 50 A

B) 62.5 A

C) 36.08 A

D) 28.87 A

Example 10-cont:

Delta-connected load, $V_L = 480\ V, P = 24\ kW, PF = 0.8\ lag.$
Find I_L

Delta-connected load, $V_L = 480\ V, P = 24\ kW, PF = 0.8\ lag.$
Find $\bar{S}$

A) $17318.4\angle 36.87^\circ$ VA

B) $30000\angle 36.87^\circ$ VA

C) $30000\angle -36.87^\circ$ VA

D) $17318.4\angle -36.87^\circ$ VA

Example 11:

$S_{tot} = 1000k$Va, PF=0.8 lag, $V_L = 4160$ V

Wye load: pure resistors, $P_1 = 200\ kW$

Delta load: impedence $\bar{Z}_\Delta$

Find $\bar{S}_2$:

Example 11-cont

$S_{tot} = 1000k$Va, PF=0.8 lag, $V_L = 4160$ V

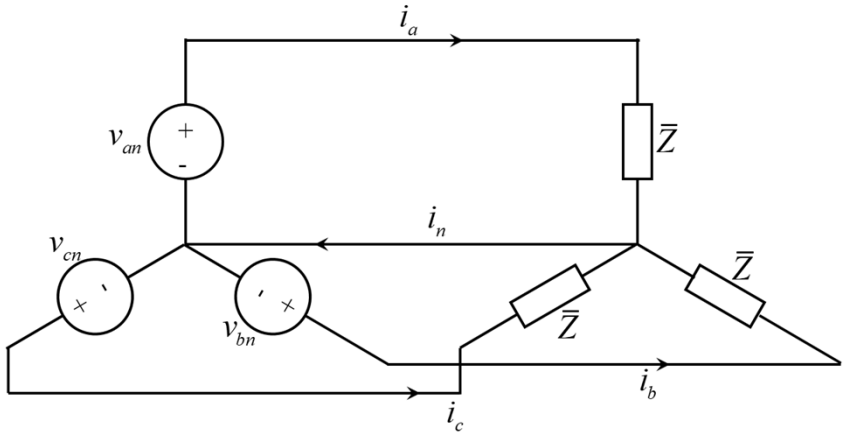
Wye load: pure resistors, $P_1 = 200\ kW$

Delta load: impedence

Find $\bar{S}_2$:

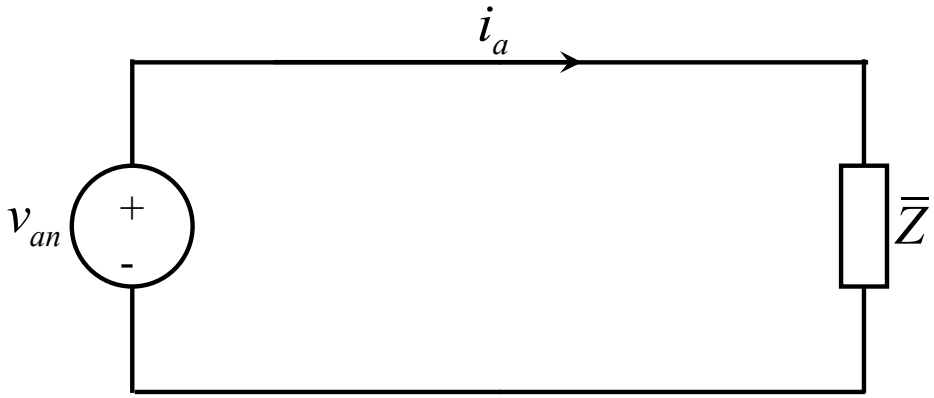
Find $\bar{Z}_\Delta$:

Per phase equivalent circuit: Wye-Wye Connection

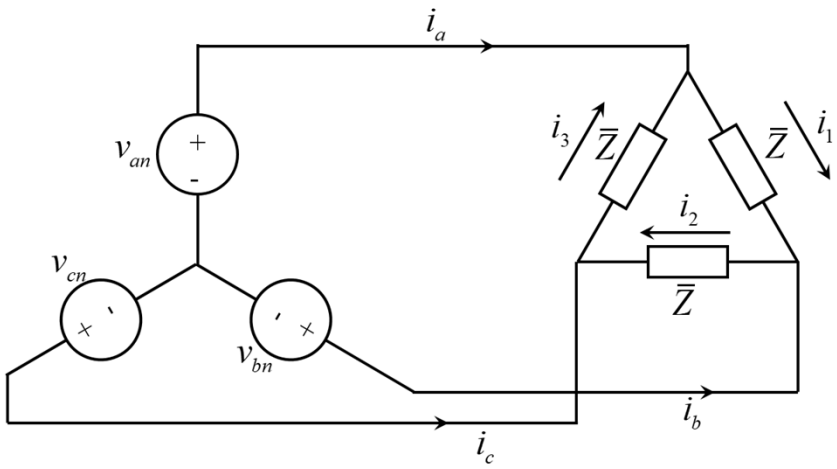


Per phase equivalent circuit

- Look only at a phase



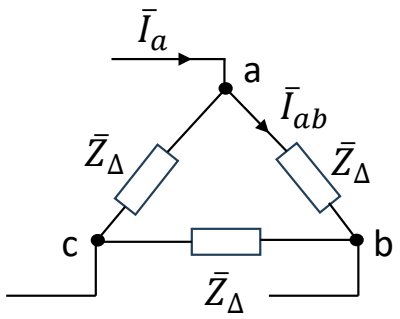
Per phase equivalent circuit: Wye-Delta Connection



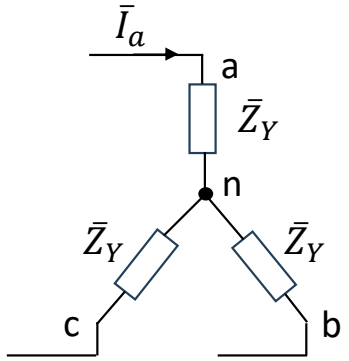
Again, only look at a phase

Delta-Wye Conversion

Delta



Wye



$$\bar{S}_{1\emptyset} = \bar{V}_{ab} \cdot \bar{I}_{ab}^{\,*} = (\bar{Z}_\Delta \cdot \bar{I}_{ab}) \cdot \bar{I}_{ab}^{\,*} = \bar{Z}_\Delta \cdot I_{ab}^{\,2} = \bar{Z}_\Delta \cdot \left(\frac{I_L}{\sqrt{3}}\right)^2$$

$\bar{Z}_\Delta \cdot \bar{I}_{ab}$ ↑

$\frac{I_a}{\sqrt{3}}$ ↑

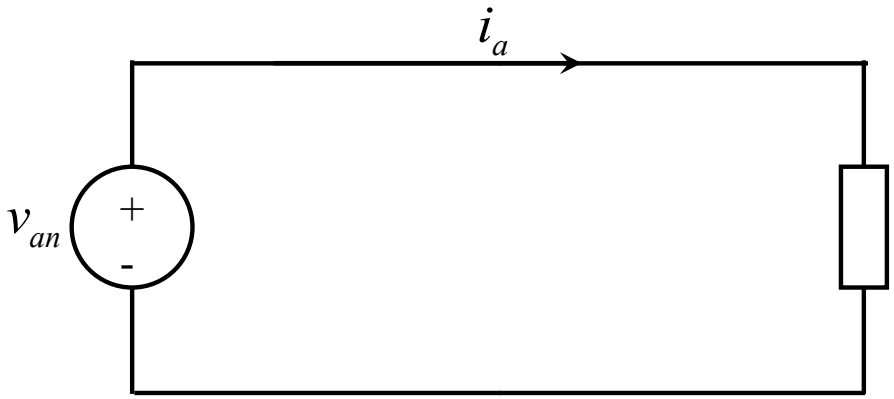
$$\bar{S}_{1\emptyset} = \bar{V}_{an} \cdot \bar{I}_a^{\,*} = (\bar{Z}_Y \cdot \bar{I}_a) \cdot \bar{I}_a^{\,*} = \bar{Z}_Y \cdot I_a^{\,2}$$

$\bar{Z}_Y \cdot \bar{I}_a$ ↑

$$\bar{S}_{1\emptyset,\Delta} = \frac{\bar{Z}_\Delta}{3} \cdot I_L^{\,2} \quad \Rightarrow \quad \bar{Z}_Y = \frac{\bar{Z}_\Delta}{3} \quad \Leftarrow \quad \bar{S}_{1\emptyset,Y} = \bar{Z}_Y \cdot I_L^{\,2}$$

To keep the same line current and complex power

Per phase equivalent circuit



Example 12:

$V_L = 208$ V, 2 Loads in Parallel

Wye load: $P_1 = 300\ kW$, PF=0.95 lag

Delta load: $\bar{Z}_\Delta = 40 + j30\ \Omega$

Find I_L, S_{tot}:

Bonus: PF correction

Given: PF_old = 0.95 lag
S_tot = 318 kVA
We need to get PF_new = 0.98 lag.
Find Q_c.